

Computational Social Science with Images and Audio

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19 September 2025

Beraja, Yang, and Yuchtman (2023)

Henceforth referred to as BYY (2023)

JOURNAL ARTICLE

Data-intensive Innovation and the State: Evidence from AI Firms in China

Martin Beraja, David Y Yang ✉, Noam Yuchtman

The Review of Economic Studies, Volume 90, Issue 4, July 2023, Pages 1701–1723,

<https://doi.org/10.1093/restud/rdac056>

Published: 13 August 2022 **Article history** ▼

BYY (2023): Motivation

Henceforth referred to as [BYY \(2023\)](#)

- ▶ AI development requires massive amounts of data
- ▶ States often collect far more data than private sector
- ▶ Government data may exceed private data in:
 - ▶ Public security
 - ▶ Defense
 - ▶ Healthcare
 - ▶ Education
 - ▶ Basic science

BYY (2023): Research question & context

- ▶ **Key Question:** Does access to government data stimulate commercial AI innovation?
- ▶ **Context:** China's facial recognition AI industry
 - ▶ Leading AI sector at technological frontier
 - ▶ Firms need large training datasets (e.g., facial video data)
 - ▶ State collects extensive surveillance data
 - ▶ Firms gain access through government contracts

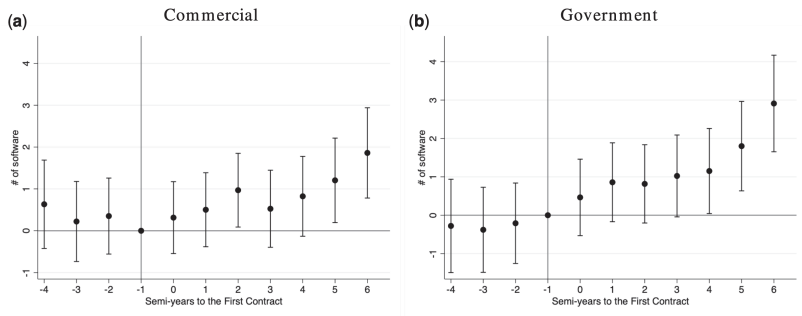
BYJ (2023): Conceptual framework

- ▶ Two types of data:
 - ▶ Government data (accessed through contracts)
 - ▶ Private sector data
- ▶ Key mechanisms:
 - ▶ Direct effect: inputs shareable across uses (data, algorithms)
 - ▶ Indirect effect: inputs not shareable across uses (e.g., learning to manage large datasets)
- ▶ Potential trade-off: Resource allocation to government projects

BYY (2023): Data & methodology

- ▶ Sample: 7,837 Chinese facial recognition AI firms
- ▶ Key measures:
 - ▶ Commercial & government software releases
 - ▶ Public security procurement contracts
 - ▶ Local surveillance capacity (cameras per capita)
- ▶ Research design:
 - ▶ Triple-differences approach
 - ▶ Compare effects of data-rich vs. data-scarce contracts

BYY (2023): Results



Source: BYY (2023). Differential cumulative software releases intended for commercial (left), for government uses (right), resulting from data-rich public security contracts, relative to data-scarce ones, controlling for firm and time period fixed effects. Data-rich contracts are defined as public security contracts in prefectures with above median surveillance capacity.

BYY (2023): Results ctd.

- ▶ Data-rich contracts lead to:
 - ▶ ~2 additional commercial software products over 3 years (cumulatively)
 - ▶ 20.2% increase relative to pre-contract level
- ▶ Concurrent increase in government software:
 - ▶ 2.9 additional products
 - ▶ 51.9% increase
- ▶ Mechanism: evidence of direct data sharing

BYY (2023): Broader implications

- ▶ Government data access can drive commercial AI innovation
- ▶ Applicable beyond facial recognition:
 - ▶ Geospatial data
 - ▶ Healthcare data
- ▶ Policy implications:
 - ▶ State procurement as innovation policy
 - ▶ Data provision strategies
 - ▶ International competitiveness (e.g., similar to discussion on privacy)
 - ▶ Important normative questions (e.g., civil liberties)
- ▶ Link to political econ debates on AI development (and innovation) in autocracies vs. democracies

Beraja, Kao, Yang, and Yuchtman (2023)

Henceforth referred to as BYKY (2023)

JOURNAL ARTICLE EDITOR'S CHOICE

AI-tocracy*

Martin Beraja, Andrew Kao, David Y Yang, Noam Yuchtman

The Quarterly Journal of Economics, Volume 138, Issue 3, August 2023, Pages 1349–1402,
<https://doi.org/10.1093/qje/qjad012>

Published: 13 March 2023 **Article history** ▼

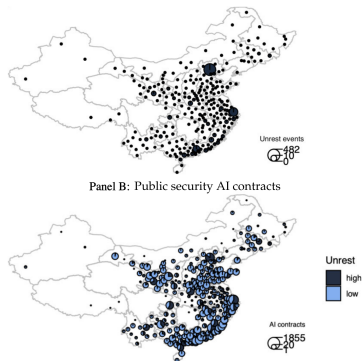
BKYY (2023): Motivation

- ▶ Traditional view: Autocracy misaligned with frontier innovation
 - ▶ Autocrats' rents eroded by technological change
 - ▶ Innovation stifled by the threat of expropriation
- ▶ However, AI may be different
 - ▶ Technology of prediction → may enhance surveillance capabilities
 - ▶ Government data valuable for development
 - ▶ Potential for commercial spillovers (see earlier paper)

BKYY (2023): Research question & context

- ▶ **Key Question:** Can autocracy and frontier AI innovation be mutually reinforcing?
- ▶ **Context:** Facial recognition AI in China (2014-2020)
 - ▶ Political control paramount for Chinese Communist Party
 - ▶ China among the world's leading producers of commercial AI innovation
 - ▶ Facial recognition relevant for surveillance/control

BKYY (2023): Research question & context ctd.



Source: BKYY (2023). Circles show prefectures with at least one first-time public security AI contract. Panel A: circle size shows no. of unrest events. Panel B: circle size shows no. of contracts (log scale); shading shows fraction of contracts procured during high-unrest periods.

BKYY (2023): Conceptual framework

Two-Way Reinforcement

- ▶ Direction 1: AI strengthens autocratic control
 - ▶ AI procurement following unrest
 - ▶ Enhanced surveillance capabilities
 - ▶ Preemptive control of dissent
- ▶ Direction 2: Political control drives innovation
 - ▶ Government contracts provide resources
 - ▶ Access to public security data
 - ▶ Spillovers to commercial applications

→ Self-reinforcing “AI-tocracy”

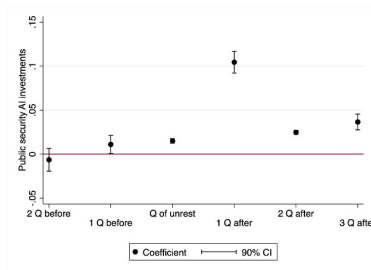
BKYY (2023): Data

- ▶ Political Unrest
 - ▶ GDELT Project database
 - ▶ Protests, demands, threats
 - ▶ Weather data for IV
- ▶ AI Development
 - ▶ Government procurement records
 - ▶ Firm software registrations
 - ▶ Export deals database
- ▶ Coverage
 - ▶ 344 Chinese prefectures
 - ▶ 2014-2020 time period
 - ▶ Comprehensive firm database

BKYY (2023): Empirical strategy

- ▶ Key Methodological Approaches:
 - ▶ Event studies around contract timing
 - ▶ IV using local weather conditions
 - ▶ Difference-in-differences comparing agencies
- ▶ Identification Challenges:
 - ▶ Endogenous unrest and procurement
 - ▶ Selection into contracts
 - ▶ Potential spillovers

BKYY (2023): Results I – Unrest drives AI procurement

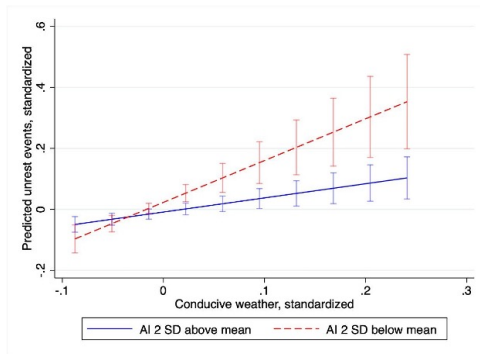


Source: BKYY (2023). This figure plots the estimated effects of leads and lags of prefecture political unrest on prefecture public security AI procurement from a regression that also includes quarter and prefecture fixed effects. The outcome (AI procurement) and explanatory variables of interest (unrest events) are standardized to mean = 0, variance = 1.

BKYY (2023): Results I – Unrest drives AI procurement ctd.

- ▶ Stronger effects using weather IV
- ▶ Complementary investments:
 - ▶ Increased camera procurement (complement)
 - ▶ Shift to high-skilled police hiring

BKYY (2023): Results II – AI effectiveness

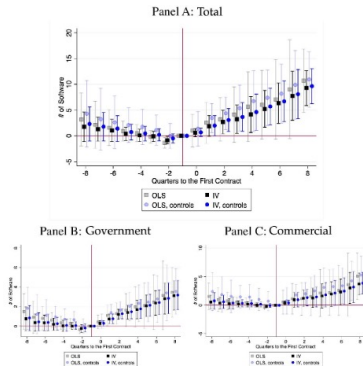


Source: BKYY (2023). Effect of conducive weather on unrest events across low and high AI adoption prefectures. The figure shows predicted unrest events given weather conditions (LASSO specification) at AI stocks 2 SD above mean (solid blue) and 2 SD below mean (dashed red), controlling for prefecture and quarter fixed effects.

BKYY (2023): Results II – AI effectiveness ctd.

- ▶ AI reduces unrest from weather shocks
 - ▶ 1 SD higher AI stock cuts effect by 25%
 - ▶ Effect doubles with camera deployment (i.e., cuts it in half)
- ▶ Effects specific to public security AI

BKYY (2023): Results III – Innovation effects



Source: BKYY (2023). Effects of initial facial-recognition AI contracts from local public security agencies during high-unrest periods on firms' total software (A), government software (B), and commercial software production (C).

BKYY (2023): Results III – Innovation effects ctd.

- ▶ Software development increases:
 - ▶ +10 products within 2 years (+48.6%)
 - ▶ Both government and commercial
- ▶ Export success:
 - ▶ 3x more likely to begin exporting
 - ▶ Indicates frontier innovation

BKYY (2023): Historical context & implications

- ▶ Similar historical episodes:
 - ▶ Soviet aerospace technology
 - ▶ Imperial Germany chemical engineering
- ▶ Modern episodes?

BKYY (2023): Conclusions

- ▶ Key Findings:
 - ▶ Mutual reinforcement between AI and autocracy
 - ▶ No evidence of innovation distortion
- ▶ Broader Implications:
 - ▶ Innovation may strengthen autocracy
 - ▶ Potential spread to other authoritarian regimes
 - ▶ Challenges assumption that democracy is necessary for frontier innovation

A counterexample: the limits of AI for authoritarian control

... and compromised AI performance; working paper by Yang (2024)

- ▶ AI may enhance autocrats' repressive capabilities (see previous discussion)
- ▶ This paper: AI effectiveness limited by existing repressive institutions
 - ▶ Strategic behavior under repression corrupts training data
 - ▶ Self-censorship reduces useful information
 - ▶ Performance worsens with repression, especially during crises

Yang (2024): In a nutshell

- ▶ Strategic behavior has two forms
 - ▶ Preference falsification: expressing false regime support
 - ▶ Self-censorship: avoiding opinion expression
- ▶ Implications for AI
 - ▶ Reduces valuable information in training
 - ▶ Makes content distinction more difficult
- ▶ Solutions?
 - ▶ More domestic data yields marginal benefit
 - ▶ International data helps but limited by different discourse